Chapter 2 : Thermochemistry

PSPM 2011 / 2012

Q. 1 (a) (i) Define standard enthalpy of combustion 1m

Answer (a) (i)

Standard enthalpy of combustion is the **energy change** / **heat released** when one mole of substance is completely burn in excess oxygen at **standard condition** / 298K & 1atm.

Q. 1 (a) (ii) The enthalpy of combustion of benzoic acid is -3226.7 kJ mol^{-1}. When 3.2g benzoic acid, C_6H_5COOH is completely combusted in a bomb calorimeter containing 2.0 kg of water, the temperature of the water increased by 3.8°C. Calculate the heat capacity of the calorimeter. 5 m

Answer (a) (ii)

Mol of C_6H_5COOH = (3.2 g / 122 g mol^{-1})
= 0.0262 mol

1 mol C_6H_5COOH $\longrightarrow$ - 3226.7 kJ
0.0262 mol C_6H_5COOH $\longrightarrow$ - 3226.7 x 0.0262
= - 84.63 kJ

$$q = mc\Delta T + C\Delta T$$
$$84.63 \text{ kJ} = (2.0 \times 10^3)(4.18)(3.8) + C(3.8)$$
$$C = 1.39 \times 10^4 \text{ J° C}^{-1}$$
$$= \mathbf{13.9\ kJ\ °C^{-1}}$$

PSPM 2012 / 2013

Q. 2 (a) Calculate the enthalpy of hydration of Cl^- in the dissolution process of LiCl in water using energy cycle method. 6m

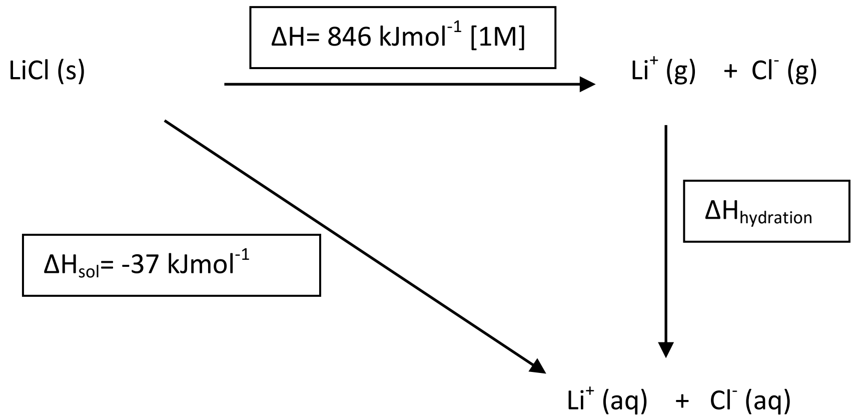
Given:

Lattice energy of LiCl = -846 $kJmol^{-1}$

Enthalpy of solution of LiCl in water = -37 $kJmol^{-1}$

Enthalpy of hydration of Li^+ = -510 $kJmol^{-1}$

Answer (a)



ΔH_{sol}= breaking lattice energy + ($\Delta H_{hydration}$ Li^+ + $\Delta H_{hydration}$ Cl^-)

-37 $kJmol^{-1}$= + 846 $kJmol^{-1}$ + (-510 +$\Delta H_{hydration}$ Cl^-)

$\Delta H_{hydration}$ Cl^-= **-373 $kJmol^{-1}$** (unit insist)

PSPM 2013 / 2014

Q. 3 (a) Solid pentaluminiumnonahydride, Al_5H_9, burns vigorously in oxygen to give solid aluminium oxide, Al_2O_3, and water, H_2O. 7 m

(i) State Hess's law.

Answer (a)(i)

The value of ΔH° for any reaction that can be written in steps equals to the sum of the values of ΔH° of each of the individual step.

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Enthalpy change in a chemical process is the same whether the process takes place in one step or several steps

(ii) By using Hess's law, calculate ΔH^{o} for the combustion of Al_5H_9.

Given:

ΔH^{o}_{f} (Al_2O_3 (s)) = -1669.8 $kJmol^{-1}$

ΔH^{o}_{f} (Al_5H_9 (s)) = -95.4 $kJmol^{-1}$

ΔH^{o}_{f} (H_2O (*l*)) = -285.8 $kJmol^{-1}$

Answer (ii)

$2Al_5H_9(s) + 12O_2(g) \rightarrow 5Al_2O_3(s) + 9H_2O(l)$

$\Delta H°_{rxn} = \Sigma\, \Delta H°_f$ (**products**) – $\Sigma\Delta H°_f$ (**reactants**)

$= [\, 5\Delta H°_f (Al_2O_3) + 9\Delta H°_f (H_2O)] – [2\Delta H°_f (Al_5H_9) + 12\Delta H°_f (O_2)]$

$= [\, 5(-1669.8) + 9(-285.8)] – [2(-95.4) + 12(0)]$

= **–10730.4 kJ**

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Several steps	$\Delta H°_f$(kJ mol^{-1})
$2Al(s) + \frac{3}{2}O_2(g) \rightarrow Al_2O_3(s)$	-1669.8 (1)
$5Al(s) + \frac{9}{2}H_2(g) \rightarrow Al_5H_9(s)$	-95.4 (2)
$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$	-285.8 (3)

$$(1)\,x\frac{5}{2} : 5Al(s) + \frac{15}{4}O_2(g) \rightarrow \frac{5}{2}Al_2O_3(s) \qquad \Delta H = -4174.5$$

$$(2) \rightleftarrows: Al_5H_9(s) \rightarrow 5Al(s) + \frac{9}{2}H_2(g) \qquad \Delta H = 95.4$$

$$(3)\,x\frac{9}{2} : \frac{9}{2}H_2(g) + \frac{9}{4}O_2(g) \rightarrow \frac{9}{2}H_2O(l) \qquad \Delta H = -1286.1$$

$$Al_5H_9(s) + 6O_2(g) \rightarrow \frac{5}{2}Al_2O_3(s) + \frac{9}{2}H_2O(l)$$

$\Delta H = – 4174.5 + 95.4 + (– 1286.1)$

$= – 5365.2$ kJ mol^{-1} or kJ

PSPM 2014 / 2015

Q. 4 (a) (i) Define lattice energy 7 m

Answer (a) (i)

Lattice energy is the standard enthalpy change when one mole of ionic solid is formed from its gaseous ions.

OR

The energy required/absorbed to completely separate one mole of solid ionic compound into gaseous ions.

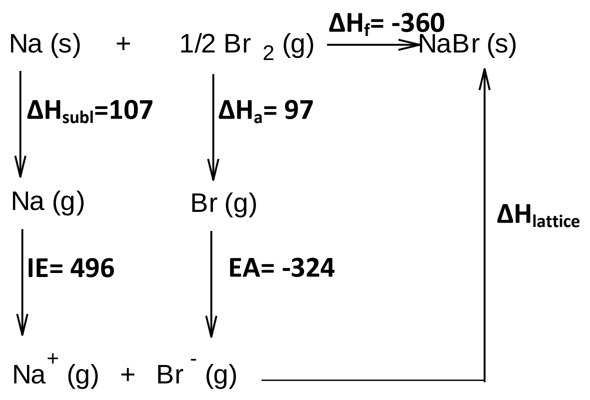
OR

The energy released when isolated gaseous ions comes together to form one mole of solid ionic compound.

Q. 4 (a)(ii) Construct a Born-Harber cycle and calculate the lattice energy for sodium bromide, NaBr, using the following data:

Enthalpy of formation of sodium bromide	= -360 $kJmol^{-1}$
Enthalpy of sublimation of sodium	= 107 $kJmol^{-1}$
Enthalpy of atomization of bromine	= 97 $kJmol^{-1}$
Ionisation energy of sodium	= 496 $kJmol^{-1}$
Electron affinity of bromine	= -324 $kJmol^{-1}$

Answer (a) (ii)

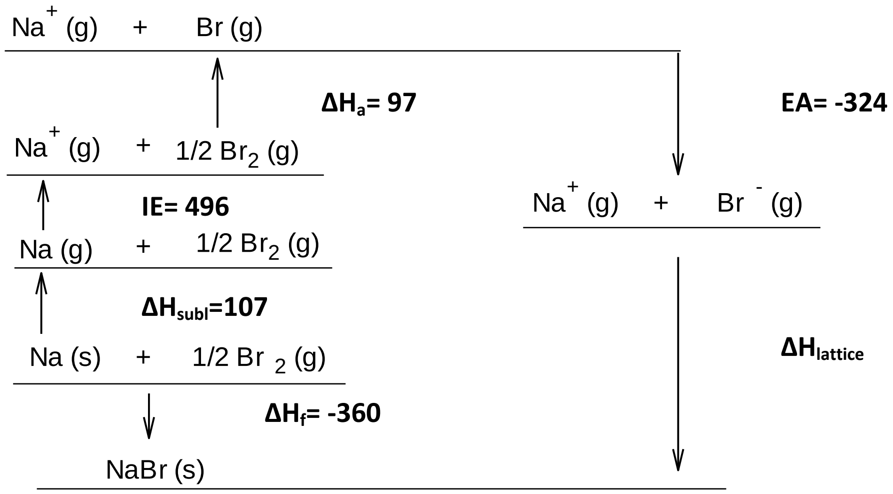


Lattice energy = ΔH_f - (ΔH_{subl} + ΔH_a + IE + EA)

= -360 – [107 + 97 + 496 + (-324)]

= **-736 $kJmol^{-1}$**

OR



Lattice energy = ΔH_f - (ΔH_{subl} + ΔH_a + IE + EA)
= -360 – [107 + 97 + 496 + (-324)]
= **-736 kJmol^{-1}**

PSPM 2015 / 2016

Q. 5 (a) A bomb calorimeter contains 780 g of water. An amount of 2.34 g benzoic acid, $C_7H_6O_2$, is burned in the calorimeter and this causes the temperature to increase from 25.6°C to 36.7°C. Calculate, 8 m

(i) energy released in kJ, if heat capacity of the empty calorimeter is 870 J°C^{-1}.

Answer (a)(i)

Heat released by reaction = Heat absorbed by water + Heat absorbed by bomb calorimeter

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Heat released by reaction, q_{rxn} = (q_{water} + q_{bomb})

q_{rxn} = mcΔT + CΔT
= (780 g x 4.18 J/g °C x 11.1 °C) + (870 J/°C x 11.1°C)
= 45847 J
= 45.85 kJ

Q. 5 (a)(ii) heat evolved per mole of benzoic acid.

Answer (a)(ii)

Mol of $C_7H_6O_2$ = 2.34 g / 122g mol^{-1}
= 0.0192

0.0192 mol $C_7H_6O_2$ released 45.9 kJ of heat
1 mol $C_7H_6O_2$ released **2390 kJ mol^{-1}**
* Enthalpy of combustion of $C_7H_6O_2$ = -2393 kJ mol^{-1}

PSPM 2016 / 2017

Q. 6 (a) The combustion of ethanol, C_2H_5OH, occurs according to the following reaction equation : 6 m

$$C_2H_5OH(l) + 3O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(g) \qquad \Delta H^{o}C = ?$$

When 3.65g sample of ethanol was burned in excess oxygen in a bomb calorimeter, the temperature rose from 25°C to 66°C.

(i) Define the standard enthalpy of combustion of ethanol.

(ii) If the heat capacity of the calorimeter is 1.98 $kJ^{o}C^{-1}$, calculate $\Delta H^{o}C$.

Answer	(a)(i)	Standard enthalpy of combustion of ethanol is the heat released when 1 mol of ethanol/compound/substances is combusted in excess oxygen at 25°C and 1 atm @ standard condition.
	(ii)	Mol of ethanol used = 3.68/46 @ 0.08 mol q = CΔT = 1.98 x (66-25) = 81.18 kJ $\Delta H^{o}_{C} = \frac{81.18\ kJ}{0.08\ mol}$ = **-1014.7 kJ/mol**

PSPM 2017 / 2018

Q. 7 (a) The standard enthalpies of formation of NO and NO_2 are 90.37 and 33.80 $kJmol^{-1}$ Respectively. 9 m

(i) Give the definition of standard enthalpy of formation

(ii) By referring to 1 (a)(i), explain why the standard enthalpy of formation for elements is zero

(iii) Write the thermochemical equations that show the formation of NO and NO_2

(iv) Determine the enthalpy change of the following reaction using an energy cycle diagram

$$2NO + O_2 \longrightarrow 2NO_2$$

Answer (a) (i) Enthalpy change / heat change when 1 mol of compound / substance is formed from its elements under standard conditions (25° C, 1 atm)

(ii) Elements are formed from their respective atoms / elements

(iii) $\frac{1}{2} N_2 (g) + \frac{1}{2} O_2 (g) \longrightarrow NO (g) \quad \Delta H = +90.37 \text{ KJ mol}^{-1}$

$\frac{1}{2} N_2 (g) + O_2 (g) \longrightarrow NO_2 (g) \quad \Delta H = +33.80 \text{ KJ mol}^{-1}$

(iv)

$$2NO (g) + O_2 (g) \longrightarrow 2NO_2 (g)$$

$$2NO (g) + O_2 (g) \xrightarrow{2 \times -90.37} N_2 (g) + 2O_2 (g) \xrightarrow{2 \times 33.80} 2NO_2 (g)$$

$\Delta H = (2 \times -90.37) + (2 \times 33.80)$

= **- 113.14 KJ**

PSPM 2018 / 2019

Q. 8 (a) In an experiment to measure the enthalpy change of neutralization, 25.0 mL of 1.0 M HCl is placed in a polystyrene cup. At the same temperature, 25.0 mL of 1.0 M NaOH is added and stirred. The temperature of the resulting solution increases by $7^{o}C$. Calculate the enthalpy of neutralization for this reaction. 6m

[Assume the density and specific heat capacity of the final solution are the same as water].

Answer

Mass solution = 1 g mL^{-1} x 50 mL
= 50 g

Heat evolved, $q = m_s c_s \Delta T$
=(50)(4.18)(7.0)
=**1463J @ 1.46kJ**

$HCl + NaOH \longrightarrow NaCl + H_2O$

Mol HCl = mol NaOH = mol H_2O

No mol of H_2O $= \frac{(1)(25)}{1000}$
= **0.025 mol**

Enthalpy of neutralization = q/n
= -1.46kJ/0.025mol
= -58.5 kJ mol $^{-1}$

(b) Thermochemical data for the formation of sodium oxide, Na_2O is given in **TABLE 2.**

TABLE 2

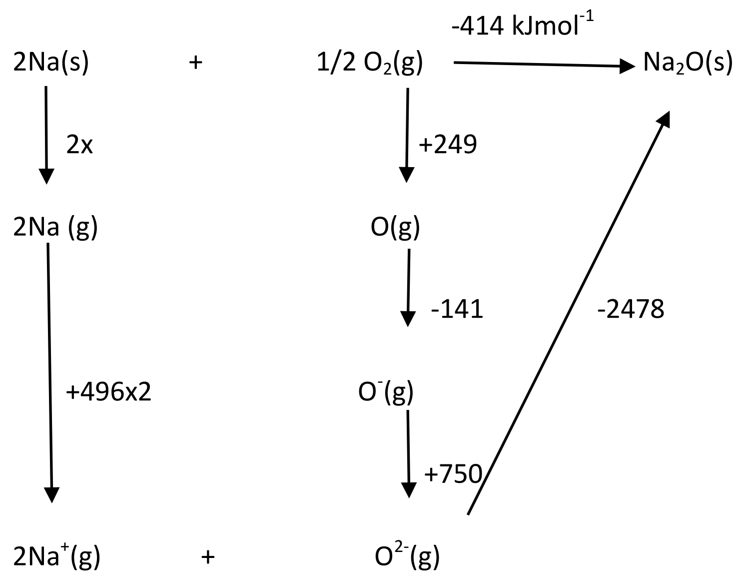
Enthalpy	**ΔH^o (kJ mol^{-1})**
Ionisation of energy of sodium	+496
Enthalpy of formation of sodium oxide	-414
First electron affinity of oxygen	-141
Second electron affinity of oxygen	+750
Enthalpy of atomization of oxygen	+249
Lattice energy of sodium oxide	-2478

8m

Based on **TABLE 2,**

(i) Construct a Born-Haber cycle for sodium oxide.

Answer



(ii) Calculate the enthalpy of atomization of sodium.

Answer

$-414 = (249 - 141 + 750 + 2x + (496x2) - 2478)$

$X = 107$ kJ mol^{-1}

$\Delta H_{atomisation}$ of sodium is +107 kJ mol^{-1}